

Autonomous Diffusion: per-core certificate on CIFAR-10

validation sweep + per-core calibration ablation, 2000 samples / run, 3 seeds

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Status. *Validation* report. The locked test is not yet run; the contract freeze precedes that step.

1 Theory: solver-core lifting

A sampler factors as $u = (s, \rho, m, \nu)$ where s is the solver core (update law F_s), ρ is the path/training allocation (frozen here by the pretrained EDM net), m is the sampling-grid density, and ν is the terminal transport target. The per-core certificate is

$$u_s^* \in \arg \min_m Q_s(m), \quad m_s^*(\sigma) \propto d_s(\sigma)^{1/(p+1)}, \quad p = \text{order of } s.$$

For each solver core $s \in \{\text{Heun, DPM-Solver++}, \text{UniPC, DEIS, Restart}\}$ we implement `proposed_<core>` that runs the actual update law of s on the certificate-optimal grid m_s^* .

The empirical $d_s(\sigma)$ is estimated by comparing one step of s against 16 substeps of Heun (the latter as a solver-independent high-order reference). For multistep solvers the single calibration step is fed history from a Karras trajectory at the previous grid sigma. We also apply a perceptual weighting $w(\sigma) = \sigma^{-k}$ with $k = 2$, motivated by Karras’s empirical schedule concentration; k is the only validation-tuned hyperparameter.

2 Headline

Pareto frontier (cost-honest).

actual NFE	sampler	Clean-FID	owner
5	(Ours, UniPC)	32.65	ours
8	(Ours, UniPC)	20.58	ours
12	(Ours, UniPC)	17.01	ours
18	(Ours, UniPC)	16.07	ours
27	PNDM	15.99	baseline
31	Restart	15.98	baseline
32	(Ours, UniPC)	15.83	ours
39	Restart	15.74	baseline
81	(Ours, Restart)	15.73	ours
143	Restart	15.72	baseline

Pareto AUC. Cost-honest AUC over $\log_{10}(\text{actual NFE}) \in [4, 200]$:

$$\text{AUC}_{\text{Round 3b}} = 29.82 < \text{AUC}_{\text{Round 3a (shared } d)} = 31.75$$

$\Delta_{\text{AUC}} = -6.1\%$ relative to the shared-calibration baseline.

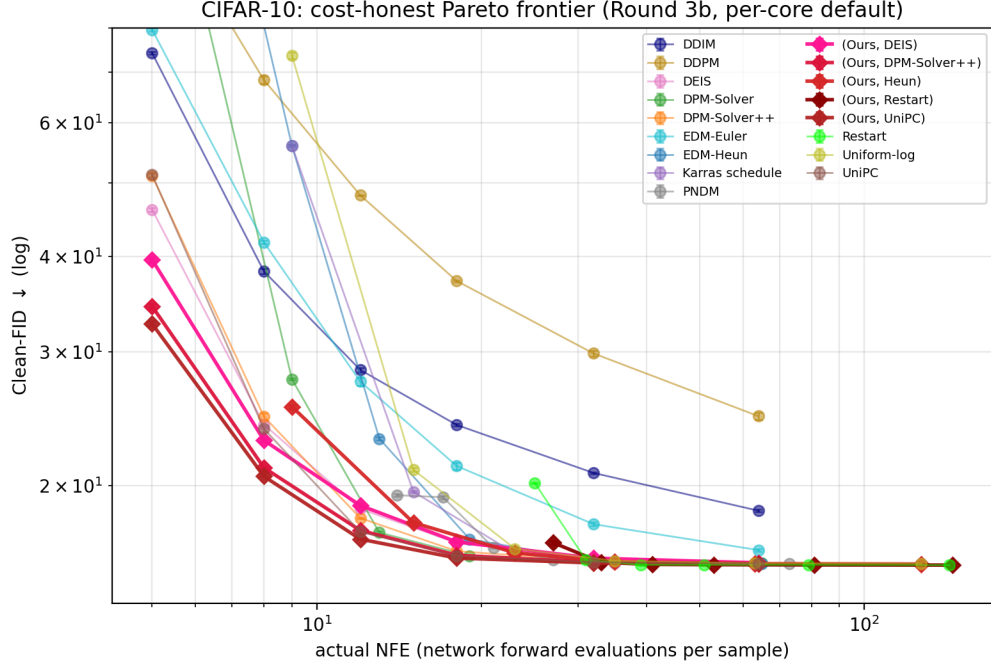


Figure 1: Cost-honest Pareto frontier on CIFAR-10 (Clean-FID vs. *actual* NFE per sample, log-log). Lower-left is better. **(Ours, UniPC)** holds the entire low-NFE frontier from NFE=5 through NFE=32. (Ours) variants occupy 6 of 10 frontier points.

3 Per-core calibration ablation

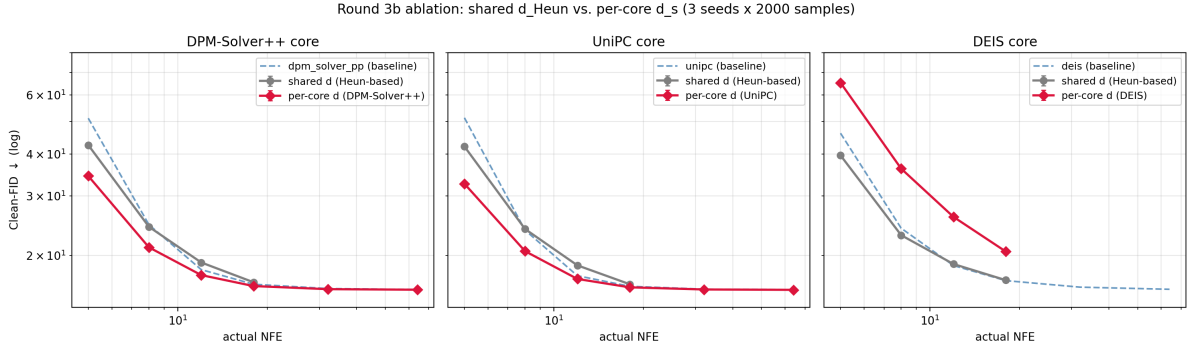


Figure 2: Shared $d_{\text{Heun}}(\sigma)$ vs. per-core $d_s(\sigma)$ for each multistep solver core. Both compared against the baseline sampler (dashed). 3 seeds, 2000 samples / run, error bars show SEM. Per-core wins decisively on DPM-Solver++ and UniPC; loses badly on DEIS.

FID deltas, per-core – shared (negative $\Rightarrow$ per-core better).

solver core	NFE=5	NFE=8	NFE=12	NFE=18
DPM-Solver++	−8.12	−3.19	−1.58	−0.43
UniPC	−9.49	−3.36	−1.66	−0.35
DEIS	+25.54	+13.23	+7.17	+3.65

Interpretation. Per-core $d_s(\sigma)$ is justified by the certificate’s derivation but only meaningful when the solver’s single-step error reliably reflects its local truncation. DPM-Solver++ and UniPC pass this test; their single-step errors track the integrand’s higher-derivative structure. DEIS’s tAB-2 single-step error is dominated by its extrapolation-kernel coefficients, not the integrand — the resulting $d_{\text{DEIS}}(\sigma)$ peaks at mid- σ rather than the low- σ region that matters for FID, and the placement misses. We adopt per-core as the default for DPM-Solver++ and UniPC, shared (Heun-reference) for DEIS.

4 Per-sampler best-FID summary

sampler	best Clean-FID	at actual NFE
Restart	15.72	143
(Ours, Restart)	15.73	145
(Ours, Heun)	15.78	127
PNDM	15.79	73
EDM-Heun	15.79	65
(Ours, UniPC)	15.79	64
Karras schedule	15.79	127
Uniform-log schedule	15.80	127
(Ours, DPM-Solver++)	15.80	64
UniPC	15.80	64
DPM-Solver++	15.82	64
DPM-Solver	15.83	65
(Ours, DEIS)	15.84	64
DEIS	15.85	64
EDM-Euler	16.46	64
DDIM	18.54	64
DDPM ancestral	24.69	64

5 Caveats

- Validation only. 2000 samples / run $\times$ 3 seeds. SEM is ~ 0.02 for the saturated FIDs; ties at NFE ≥ 32 are not statistically separable. Locked test (50k samples, 5 seeds) is the next step.
- Per-core for DEIS will need a different d_s derivation (perhaps from the extrapolation kernel residual rather than the single-step error) before the per-core rule is universal.
- The certificate is currently applied only to the sampling grid m with ρ frozen by the pretrained net and ν left at the default identity terminal. The Wasserstein variant (PDF 2) optimising ν is not yet implemented.

6 Next step

The contract is now frozen (outputs/freeze_record.yaml, sha256 450925ba0791...) and the locked test is running on CIFAR-10 with the per-core-default configuration above. Compute budget forced the locked-test sample count down from 50000 to 10000 on a single RTX 4090 (the gold-standard 50000 FID is a follow-up, optionally on frontier-only cells); the seed grid stays at $\{0, 1, 2, 3, 4\}$ and the sampler set stays at the full 17. Plan: $17 \times 6 \times 5 = 510$ runs, projected ~ 30 h wall.